

Refactor — Split Your Game into Modules

Day 01

Lesson 01

Nora XIE

Course 02

Phase 1 — Game Studio

Time: 60–75 min

Mode: Self-study

Today's goals:

- ★ Move code from one big file into modules
- ★ Use import to connect files
- ★ Keep the game playable after each move



Course 02 quest map: Day 01 of 18

Hook

Your Course 01 dragon fight probably lives in one file. Split it: `game.py` + `player.py` + `enemies.py` + `main.py`.

Learn



Learn

Module = one `.py` file. **import** = use code from another file.

After every small move, run the game. If it breaks, undo one step.

Files: `player.py` (HP, defend), `enemies.py` (dragon), `game.py` (loop), `main.py` (start).



AI rules today

Predict/Run: AI off. **Investigate:** Socratic hints only. **Make:** you review every line.

Predict

Run

Investigate

Modify

Make



Demo code

Folder: day01/

```
1 """Tiny multi-file pattern  expand this idea."""
2 from player import Player
3 from enemies import Dragon
4
5 def main():
6     hero = Player(name="Nora", hp=20)
7     boss = Dragon(hp=15)
8     print(hero)
9     print(boss)
10
11 if __name__ == "__main__":
12     main()
```

 Try it

In terminal, `cd course02/day01`, then predict output, then run: `python demos/demo_import_game.py`

 Assignment

Work in: `day01/assignment/`

A — Predict If `player.py` defines `hp = 10` but `game.py` never imports it, can `game.py` see `hp`? (yes/no + why)

B — Practice Open `assignment/monolith_dragon.py`. Copy it into your own files — do **not** delete the original until the split works.

C — Level up Split into at least `main.py` + two other modules. Game must still run `fight`, `defend`, `quit`.

D — Explain Which file would you open first to change dragon damage? Why?

 Checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest part today?
2. One line you can teach a friend.

Tomorrow: Classes and dataclasses for `Player` and `Enemy`.